

Whether historical rent trends can be used to understand and predict future rental prices in New York City

Tracy Chu, Lylian Li

I. Introduction

1. Project Aims

As study-away students living in New York, both Tracy and Lylian have directly experienced how expensive and unpredictable the city's rental market can be. Since we are about to leave New York soon, we started with a simple but meaningful question: **if we came back one year from now, how much would the apartments we rent today cost?** This personal concern inspired our project: **whether historical rent trends can be used to understand and predict future rental prices in New York City.**

To answer this question, we built a data-driven rental price study using historical Zillow rent data. We first explore rent patterns across boroughs, ZIP codes, and time, then use these findings to create features such as lagged rent, rent growth, borough indicators, and time variables for later modeling. Overall, this project aims to show how past rent trends can help explain New York's rental market and support future rent prediction.

2. Data Sources: Zillow ZORI and Reddit

This project mainly uses **Zillow Observed Rent Index (ZORI)** data from Zillow Research. Zillow is one of the largest real estate and rental platforms in the United States, and its research datasets provide large-scale housing and rental market information. ZORI measures the typical observed market-rate rent in a given region, making it useful for comparing monthly rent trends across New York ZIP codes.

ZORI is a repeat-rent index that measures the typical observed market-rate rent in a given region. It is important to note that ZORI is **not the listed price of one specific apartment** on rental platforms. Since it reflects broader ZIP-code-level rent trends across housing units and property types, its values may look lower than the rents commonly seen on rental websites, especially for popular or luxury apartments in Manhattan. However, this also makes ZORI useful for our project because it provides a consistent, comparable, monthly measure of rent across ZIP codes.

In addition to Zillow data, we also collect Reddit posts from NYC-related subreddits to add a social perspective to the analysis. Rent is not only a market number but also a real living experience. Therefore, Reddit discussions help us observe whether online complaints about rent and housing affordability increased during the same period when rent trends changed sharply. Reddit is used as a supplementary source rather than the main modeling dataset.

II. Data Description and Data Preparation

1. Access Dataset and Build DataFrame

2. Data Cleaning

For this project, we narrow the dataset to New York City ZIP codes. We keep ZIP codes located in the five NYC counties:

- New York County = Manhattan
- Kings County = Brooklyn
- Queens County = Queens
- Bronx County = Bronx
- Richmond County = Staten Island

After filtering, the NYC dataset contains **152 ZIP codes**. We then add a borough variable based on the county name.

(152, 144)

RegionID	SizeRank	RegionName	RegionType	StateName	State	City	Metro	CountyName	2015-01-31	...	2025-06-30	2025-07-31	2025-08-31	2025-09-30	2025-10-31	2025-11-30	2025-12-31	2026-01-31	2026-02-28	2026-03-31
3	62080	4	11368	zip	NY	NY New York	New York-Newark-Jersey City, NY-NJ-PA	Queens County	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2700.000000
6	62093	7	11385	zip	NY	NY New York	New York-Newark-Jersey City, NY-NJ-PA	Queens County	NaN	...	3044.561878	3110.006786	3089.504783	3071.414298	3031.055703	3014.708892	2973.686373	3010.333460	3059.504855	3110.629630
8	62019	9	11208	zip	NY	NY New York	New York-Newark-Jersey City, NY-NJ-PA	Kings County	NaN	...	2845.342594	2909.551010	3016.331449	3072.160061	3155.939273	3205.441432	3065.921897	2976.415274	3017.804947	3076.214286
15	62046	16	11236	zip	NY	NY New York	New York-Newark-Jersey City, NY-NJ-PA	Kings County	NaN	...	NaN	NaN	2857.114308	2958.952092	3128.457579	3180.443234	3064.320348	3156.161825	3143.126917	3162.500000
16	61807	17	10467	zip	NY	NY New York	New York-Newark-Jersey City, NY-NJ-PA	Bronx County	NaN	...	2416.599551	2534.798815	2606.677096	2530.246942	2581.748354	2558.887553	2569.694683	2487.993981	2545.825631	2629.875000

5 rows × 144 columns

3. Feature Engineering

To make the data suitable for time-series analysis and predictive modeling, we transform the dataset from wide format to long format. After this transformation, each observation represents **one ZIP code in one month**, with one corresponding ZORI rent value.

The main variables used in the analysis include:

- **zip_code**: ZIP code of the rental market area
- **borough**: NYC borough
- **date**: monthly observation date
- **year / month / quarter**: time features
- **zori_rent**: current monthly Zillow Observed Rent Index
- **rent_lag_1 / rent_lag_2 / rent_lag_3 / rent_lag_12**: previous rent values from 1, 2, 3, and 12 months ago
- **rent_growth_1m**: one-month rent growth rate
- **rent_growth_12m**: year-over-year rent growth rate
- **target_next_month_rent**: next month's rent index, used as the regression target
- **is_lic**: indicator for Long Island City ZIP codes, including 11101 and 11109

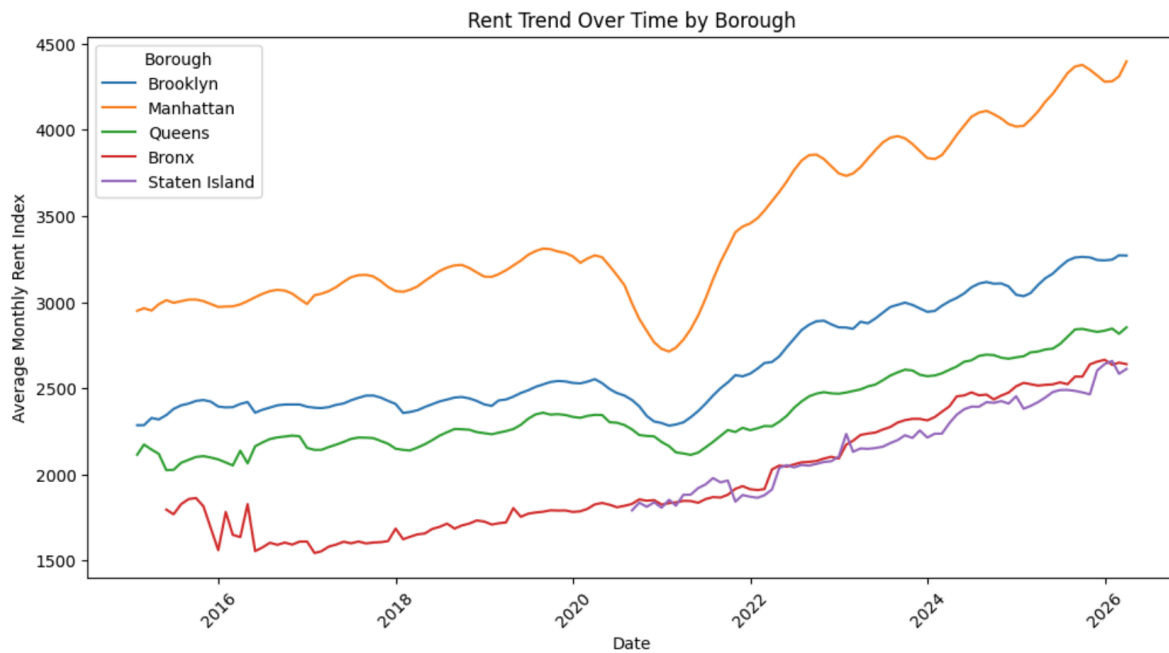
(10849, 17)

	zip_code	borough	city	county	date	year	month	quarter	zori_rent	rent_lag_1	rent_lag_2	rent_lag_3	rent_lag_12	rent_growth_1m	rent_growth_12m	target_next_month_rent	is_lic
1947	10001	Manhattan	New York	New York County	2016-01-31	2016	1	1	4048.999810	4076.478274	4084.053297	4136.821172	3958.238156	-0.006741	0.022930	4041.609677	0
2099	10001	Manhattan	New York	New York County	2016-02-29	2016	2	1	4041.609677	4048.999810	4076.478274	4084.053297	3989.656803	-0.001825	0.013022	4056.968346	0
2251	10001	Manhattan	New York	New York County	2016-03-31	2016	3	1	4056.968346	4041.609677	4048.999810	4076.478274	3996.446526	0.003800	0.015144	4087.752041	0
2403	10001	Manhattan	New York	New York County	2016-04-30	2016	4	2	4087.752041	4056.968346	4041.609677	4048.999810	4073.226094	0.007588	0.003566	4139.052840	0
2555	10001	Manhattan	New York	New York County	2016-05-31	2016	5	2	4139.052840	4087.752041	4056.968346	4041.609677	4143.042151	0.012550	-0.000963	4150.624433	0

III. Exploratory Data Analysis

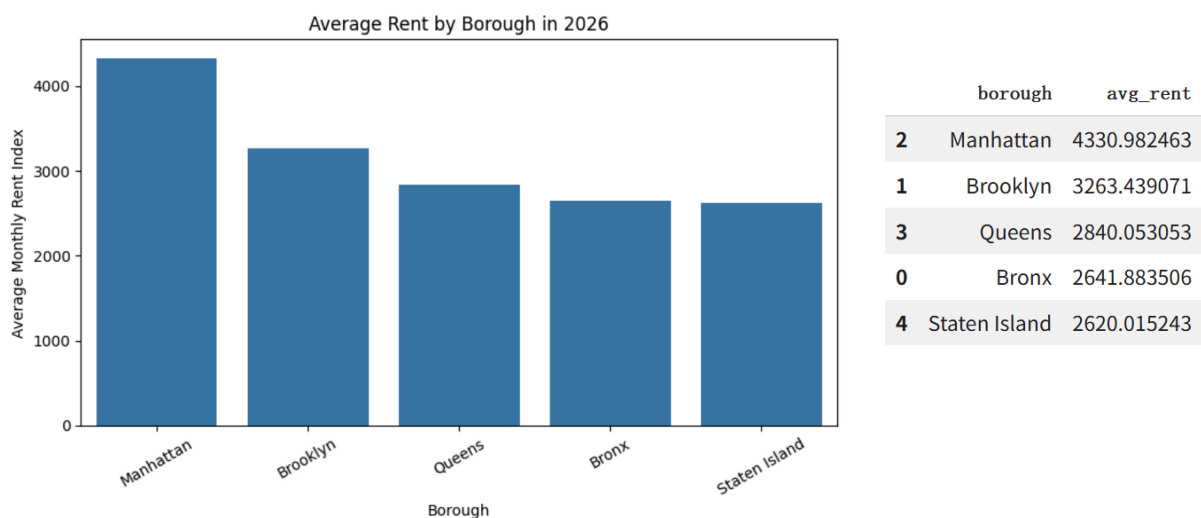
1. Rent Trend over Time by Borough

The rent trend shows that Manhattan consistently has the highest rent level, followed by Brooklyn and Queens. All boroughs experienced a clear rent drop around 2020–2021, likely reflecting the pandemic-period rental market shock, followed by a strong recovery and continued growth after 2021.



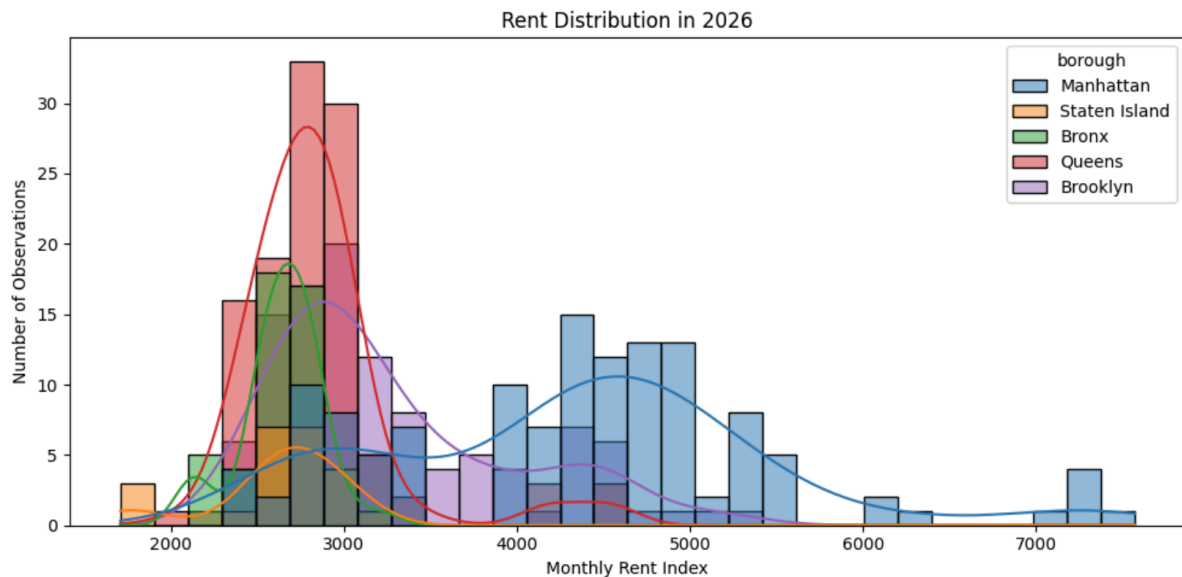
2. Average rent by borough in 2026

In 2026, Manhattan has the highest average ZORI rent among the five boroughs, at about \$4,331. Brooklyn is the second highest, around \$3,263, followed by Queens at about \$2,840. The Bronx and Staten Island have the lowest average rents, both around \$2,600. Overall, the result shows a clear rent gap between Manhattan and the other boroughs, while Brooklyn and Queens form a middle-rent group and the Bronx/Staten Island remain relatively more affordable.



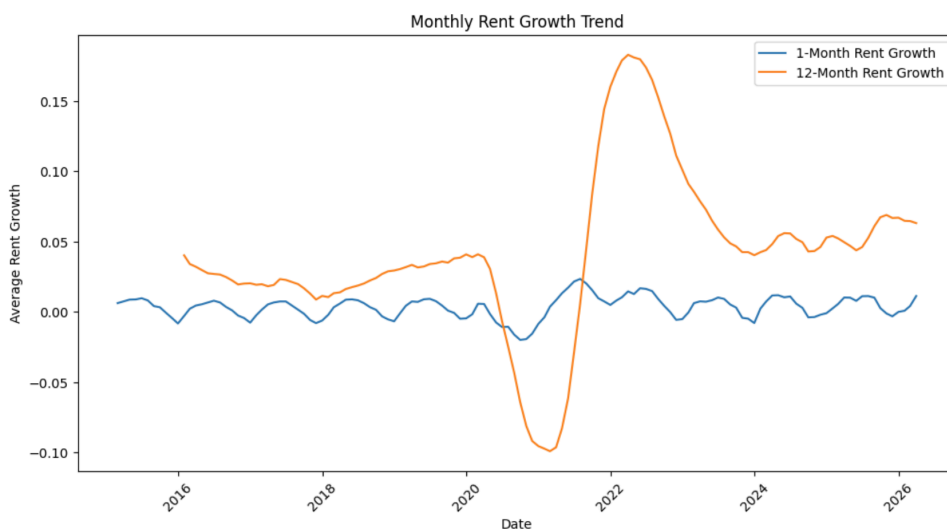
3. Rent distribution in 2026

The 2026 rent distribution shows that Manhattan has both the highest rent level and the widest spread. Overall, the distribution confirms that Manhattan is the most expensive and least homogeneous borough in terms of rent.



4. Monthly rent growth trend

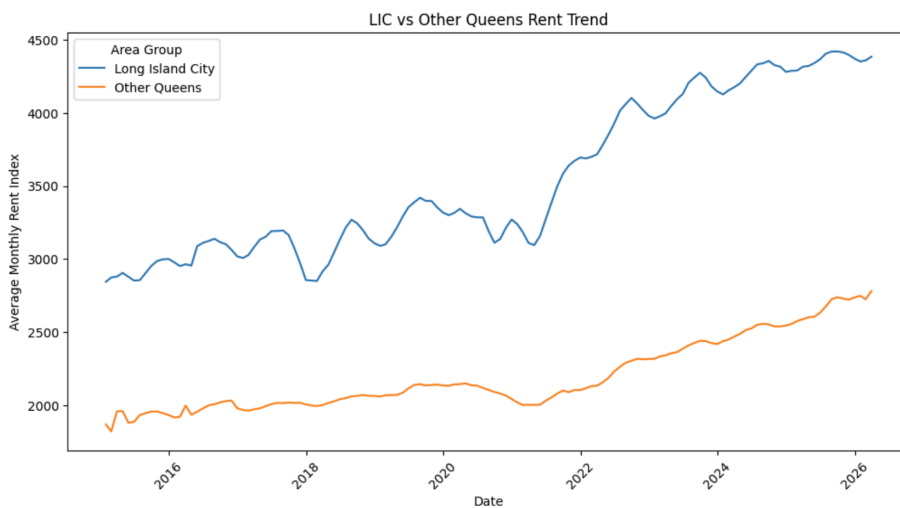
Rent growth dropped sharply around 2020, especially in the 12-month growth rate, reflecting the COVID-era rental market shock. After 2021, rent growth rebounded quickly and peaked around 2022. Since then, growth has slowed but remained mostly positive through 2026.



	date	avg_rent_growth_1m	avg_rent_growth_12m
132	2026-01-31	0.000869	0.064785
133	2026-02-28	0.004220	0.064610
134	2026-03-31	0.011344	0.063161

5. LIC vs Other Queens rent trend

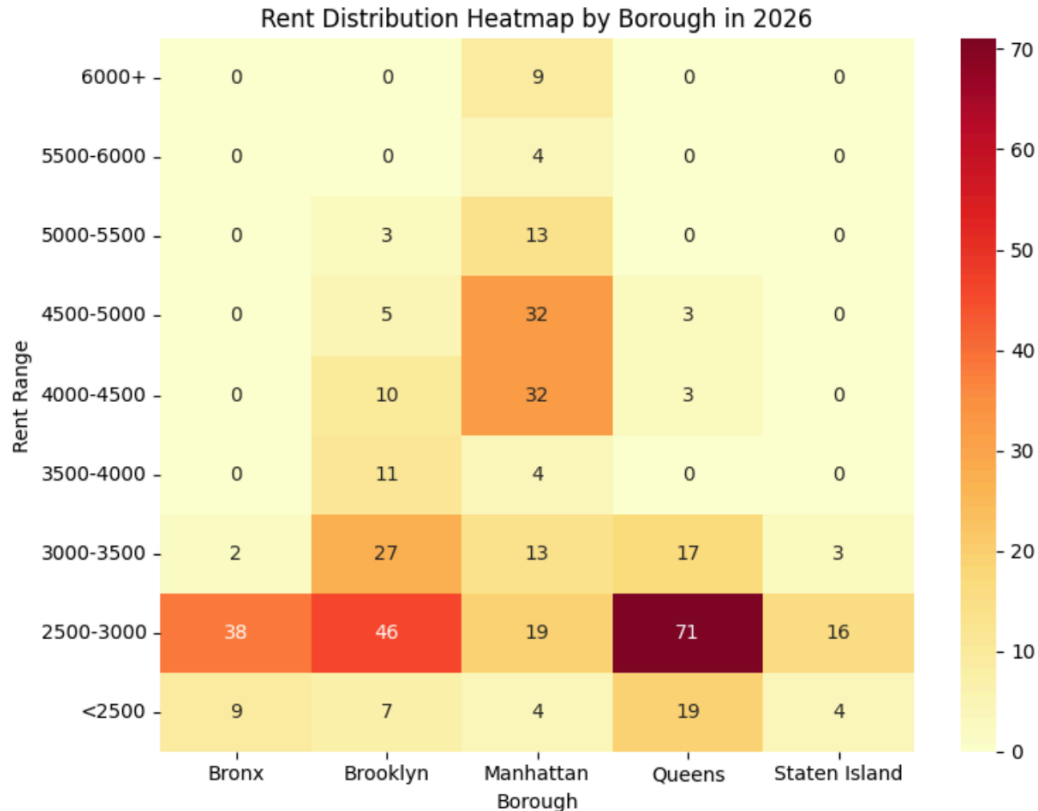
Long Island City has consistently higher rent than the rest of Queens. After a short decline around 2020–2021, LIC rent rebounded strongly and reached about \$4,300–\$4,400 by early 2026. Other Queens also increased, but much more gradually, staying around \$2,700.



	date	area_group	avg_rent
264	2026-01-31	Long Island City	4350.075563
265	2026-01-31	Other Queens	2749.774112
266	2026-02-28	Long Island City	4357.416104
267	2026-02-28	Other Queens	2725.995268
268	2026-03-31	Long Island City	4382.671403

6. Rent distribution heatmap by borough in 2026

This heatmap shows how 2026 rent observations are distributed across rent ranges in each borough. Manhattan has more observations in the higher rent ranges, while Queens, the Bronx, and Staten Island are mostly concentrated in lower rent ranges. This confirms that rent differences are not only about average rent, but also about how rent levels are distributed across areas.



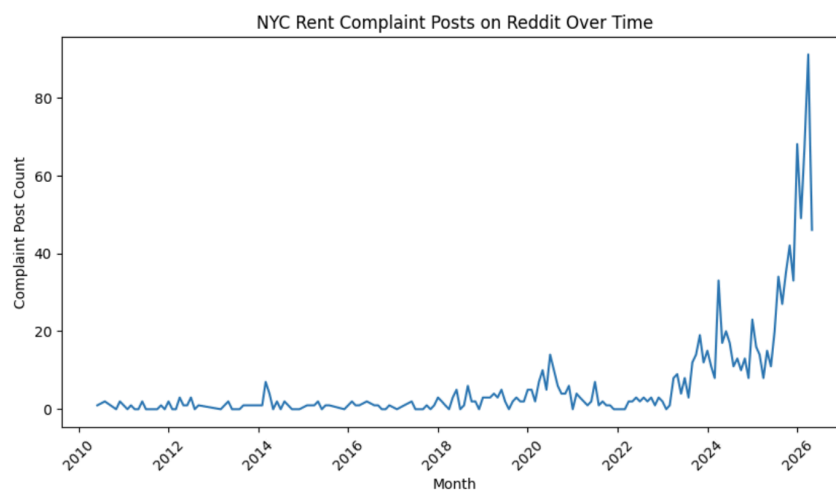
7. Social Media Complaint Analysis

To add social context to the Zillow rent index data, we collect Reddit posts from NYC-related subreddits using rent-related search terms. We classify posts as complaints if they contain keywords such as “expensive,” “unaffordable,” “priced out,” or “too high.” We then aggregate the results by month to compare online complaint intensity with rental market trends.

- Data source: Reddit posts
- Unit: post
- Time level: month or year
- Purpose: measure rent complaint frequency and intensity

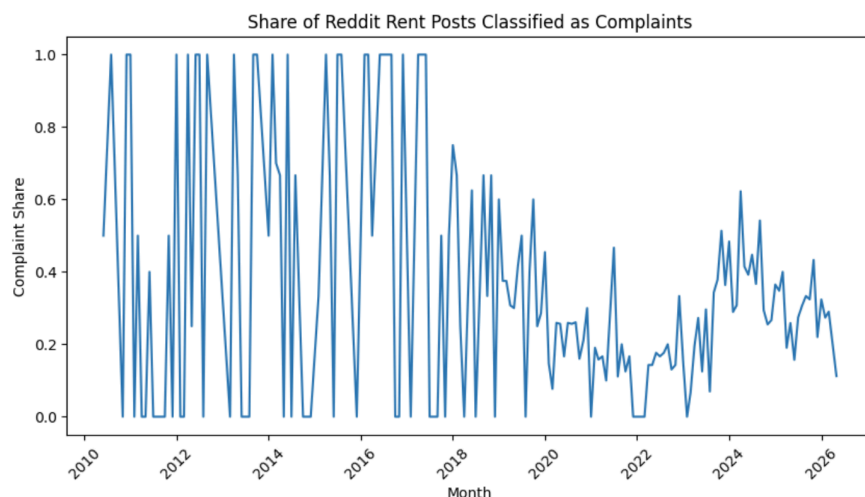
Monthly complaint count

Reddit rent complaint posts stay very low before 2020, then rise sharply after 2022 and peak around 2025–2026, suggesting that online discussion about NYC rent became much more active in recent years.



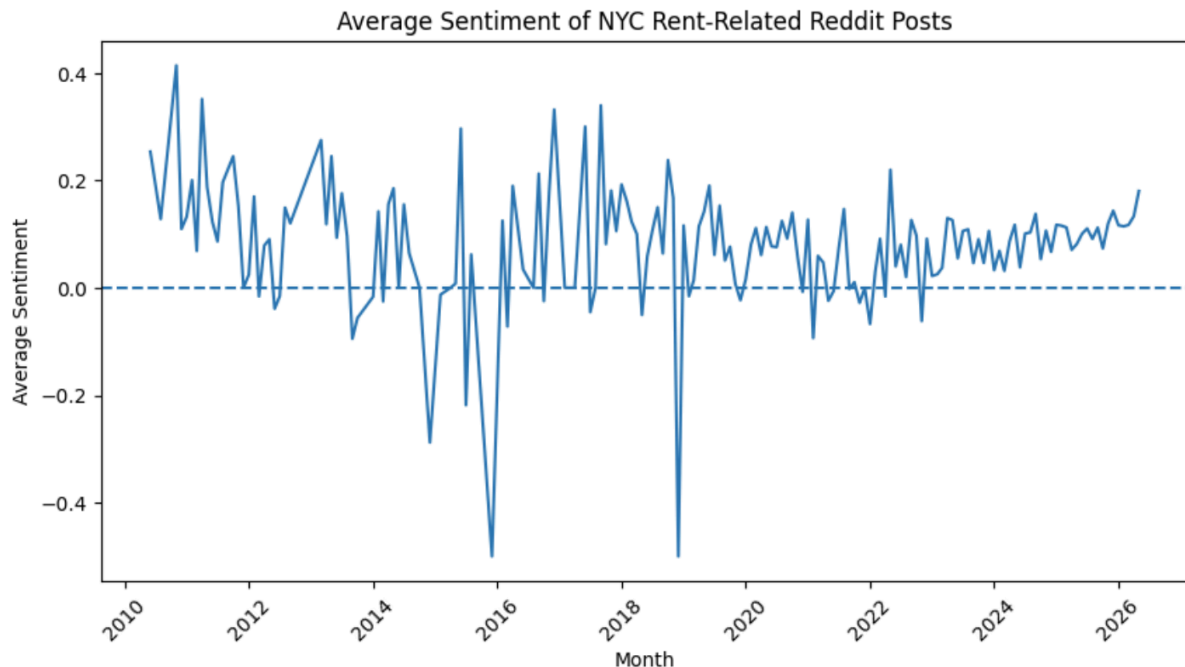
Complaint share over time

The complaint share is very unstable in earlier years because there were few posts, but after 2022 it becomes more meaningful and generally shows that a sizable portion of rent-related posts are complaints.



Average sentiment over time

A lower score means posts are more negative on average. Average sentiment stays mostly near or slightly above zero, with occasional negative spikes, suggesting that TextBlob sentiment does not capture rent frustration as clearly as the keyword-based complaint count



After visualizing all those data, we proposed these **hypothesis**:

- Manhattan has the highest rent level.
- LIC has higher rent than many other Queens ZIP codes.
- Recent rent values are strong predictors of next month's rent.
- 12-month lag captures yearly seasonality.
- Models with lag features should outperform a simple baseline.

IV. Regression modeling

1. Baseline model

A model must have an MAE lower than about \$786 to be better than simply predicting the average rent for every observation. Because RMSE is much larger than MAE, it suggests some test observations have very large errors, probably expensive ZIP codes or high-rent months.

The dummy model performs especially **poorly** on ZIP codes with **rents far from the citywide average**, such as high-rent Manhattan or lower-rent outer-borough areas.

2. Linear Regression

Linear regression gives an interpretable baseline model. It assumes a linear relationship between lagged rent features, time variables, borough, and next month's rent.

3. Ridge Regression

Ridge is good when lag variables are highly correlated. Ridge regression adds regularization, which reduces overfitting and helps when predictors such as current rent and lagged rent are strongly correlated.

4. KNN Regression

KNN predicts rent by looking at similar ZIP-code-month observations. Because KNN is distance-based, scaling is necessary.

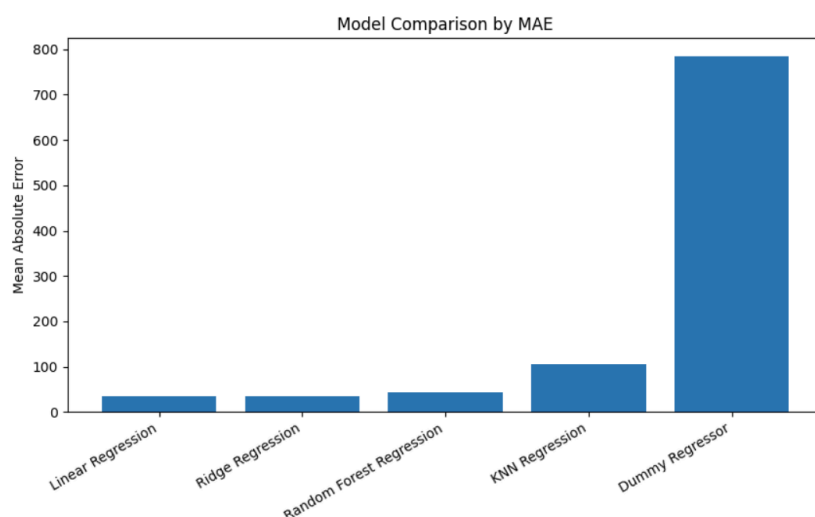
5. Random Forest Regression

Random Forest can capture nonlinear relationships, but it is less interpretable than linear regression. It may perform better if rent trends differ across boroughs or ZIP codes in complex ways.

6. Model Comparison

The strong performance of **Linear Regression** is closely related to the **time-series structure** of the Zillow ZORI dataset. Since the target variable is next month's rent and the model includes **current and lagged rent values**, the prediction problem contains strong autocorrelation. In other words, rent in one month is highly correlated with rent in nearby months. Because rent indices usually change gradually, next month's rent can be approximated well as a linear function of recent rent levels and rent growth. Therefore, the linear model performs better than more complex models such as Random Forest and KNN. However, this should not be interpreted as a causal relationship; the model is identifying **predictive temporal patterns**, not proving that one variable causes another.

MEA, RMEA, and R² Comparison

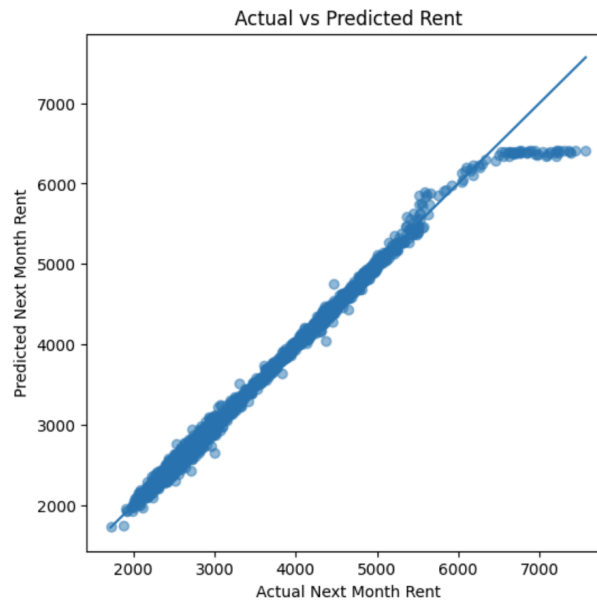


	Model	MAE	RMSE	R2
1	Linear Regression	33.904391	47.668568	0.997795
2	Ridge Regression	34.495342	47.747868	0.997787
4	Random Forest Regression	43.247877	90.275702	0.992090
3	KNN Regression	104.743392	151.397727	0.977753
0	Dummy Regressor	785.577186	1139.471460	-0.260201

Random Forest Prediction Accuracy

The actual-versus-predicted plot shows that the Random Forest model performs well for most observations, with points closely following the diagonal reference line. However, **the model underpredicts very high-rent ZIP-code-month observations**, as shown by the **flattened**

prediction pattern at the top-right of the graph. This helps explain why Random Forest has a



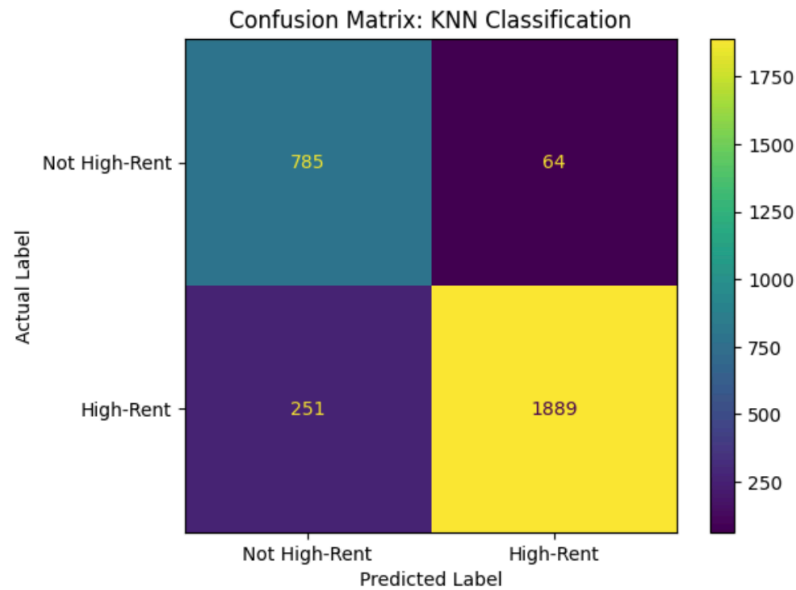
higher RMSE than Linear Regression, despite still achieving a strong overall R^2 .

V. Classification Model

1. Logistic Regression classification

Logistic Regression estimates whether next month's rent will be above the high-rent cutoff. It is useful because it gives an interpretable classification model.

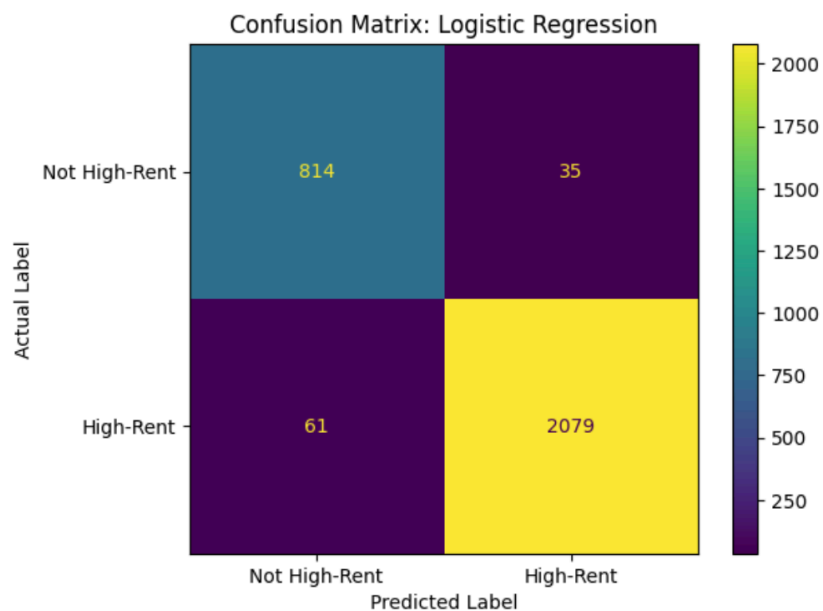
Logistic Regression performs best in the classification task, with an **accuracy of 96.8%** and an F1 score of 0.977. The confusion matrix shows that **the model correctly identifies most high-rent ZIP-code-month observations**, with only 61 false negatives out of 2,140 actual high-rent cases.



2. KNN Classification

KNN Classification predicts whether a ZIP-code-month is high-rent by looking at similar observations in the training data.

KNN Classification performs reasonably well, with an **accuracy of 89.5%**, but it produces **more false negatives and has a lower recall**.



3. Classification comparison table

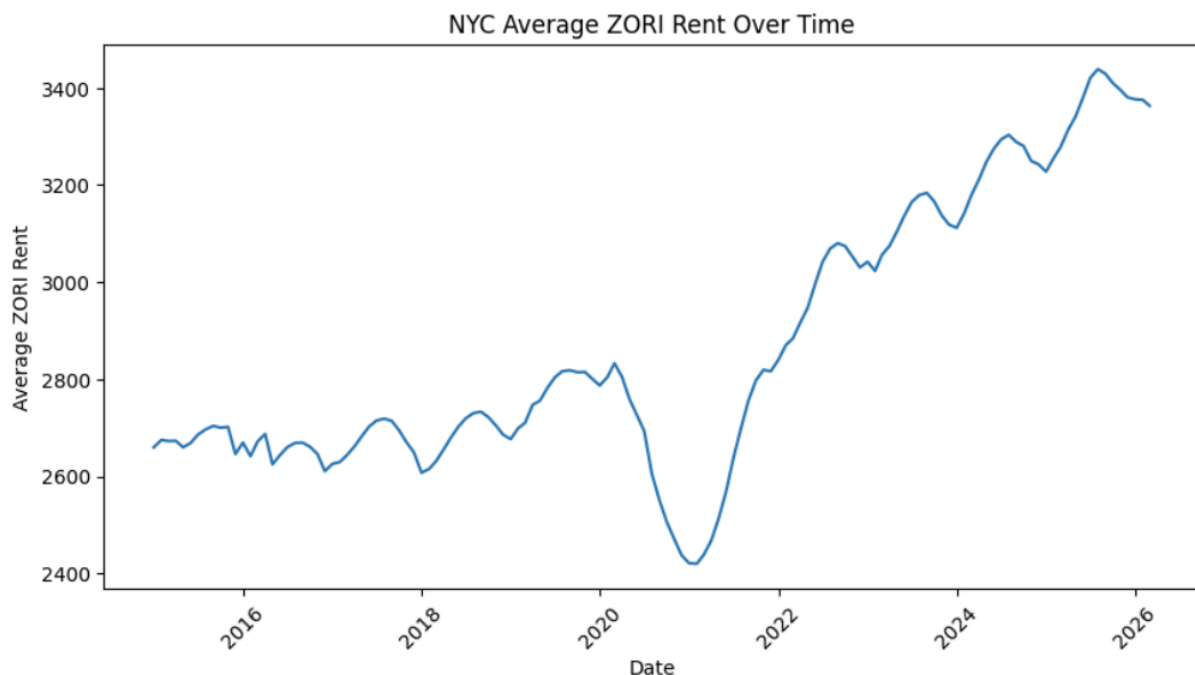
The result suggests that **the high-rent classification boundary is relatively linear** and can be captured effectively by **Logistic Regression** using current rent, lagged rent, rent growth, time, and borough features.

	Model	Accuracy	Precision	Recall	F1
0	Logistic Regression	0.967882	0.983444	0.971495	0.977433
1	KNN Classification	0.894614	0.967230	0.882710	0.923039

VI. Prediction Model

In the forecasting extension, we compare a naive baseline, ARIMA, and Exponential Smoothing using NYC average monthly ZORI rent. The **naive model** performs worst, with an MAE of about 123, because it does not adjust well to the recent high-rent period. **ARIMA** improves substantially, reducing MAE to about 37, showing that the **time-series structure** contains useful forecasting information. **Exponential Smoothing** performs best, with an MAE of about 23 and RMSE of about 29. This suggests that **NYC rent trends are smooth** and recent observations are especially informative for short-term forecasting.

1. Plot original series



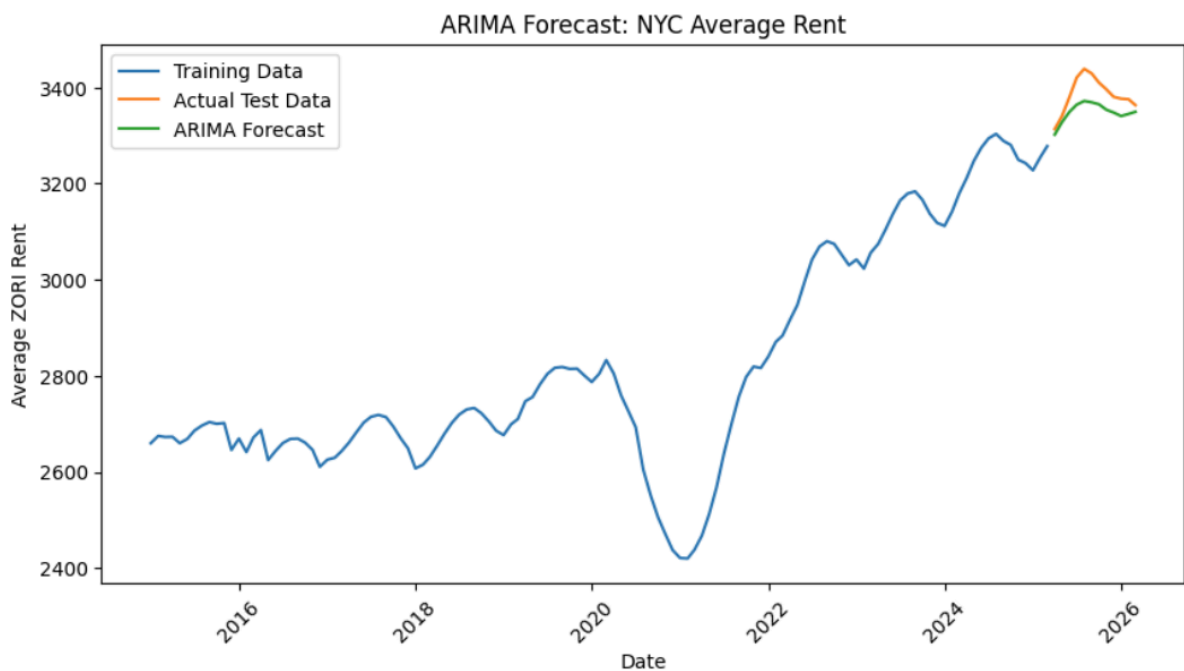
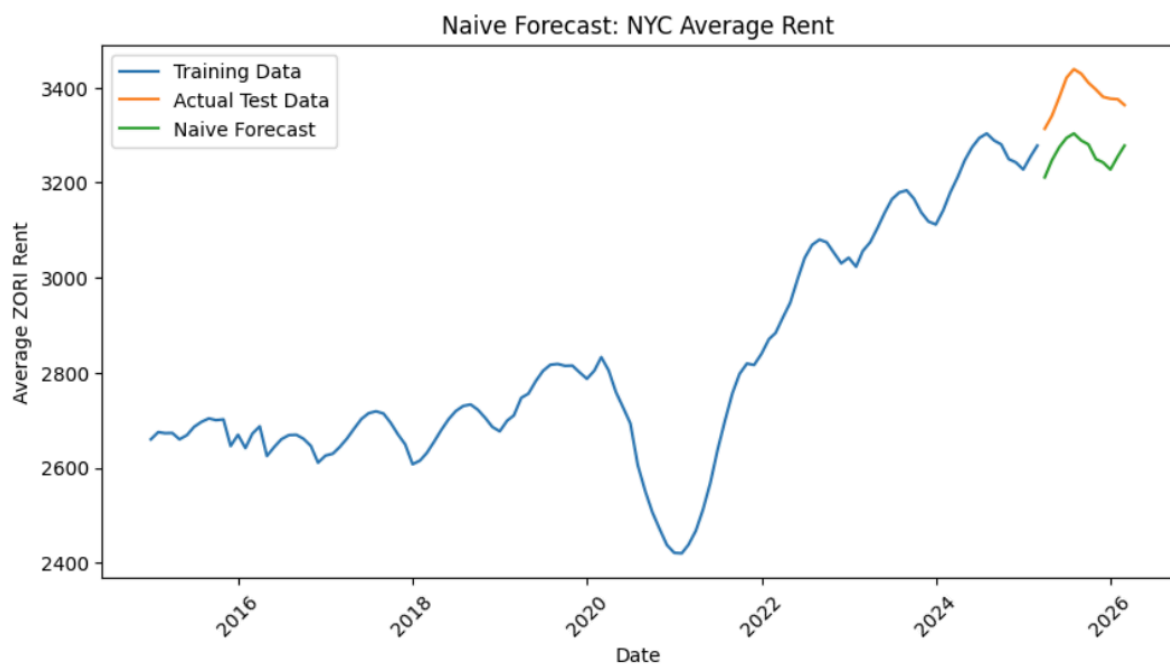
2. Create forecasting horizon

Model 1: Naive Forecasting Baseline

The Naive Forecast is much **less accurate** because it mostly repeats past patterns and **does not adjust well to the recent rent level**. In the plots, the naive line stays too low compared with the actual test rent.

Model 2: ARIMA Forecasting

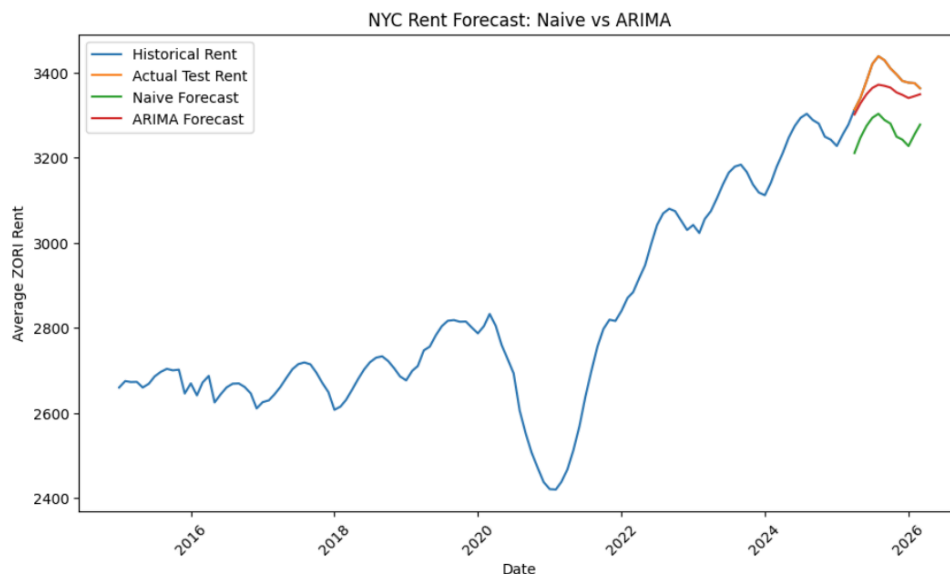
ARIMA tracks the test period **much better**. It captures the general trend and **stays close to actual rent**, but it slightly **underpredicts some of the higher values**.



6. Compare Naive vs ARIMA

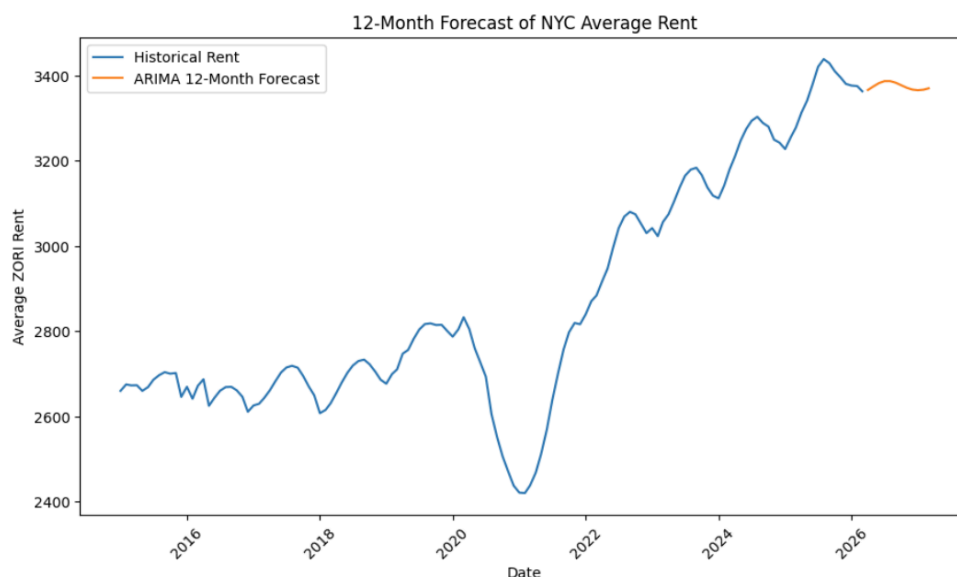
	Model	MAE	RMSE
0	Naive Forecast	122.615109	124.307401
1	ARIMA	36.591826	40.664713

In this plot, the historical rent series shows a clear **COVID-era drop** around 2020–2021, **followed by a strong recovery and upward trend after 2021**. During the test period, ARIMA is much closer to actual rent than the naive forecast, while the naive forecast **consistently underpredicts**.



7. Forecast future 12 months

In the 12-month forecast plot, **ARIMA** predicts that NYC average rent will **remain high but relatively stable**, staying around the upper **\$3,300 range** rather than continuing to rise sharply. This suggests the **model expects slower rent growth** after the recent post-pandemic increase.



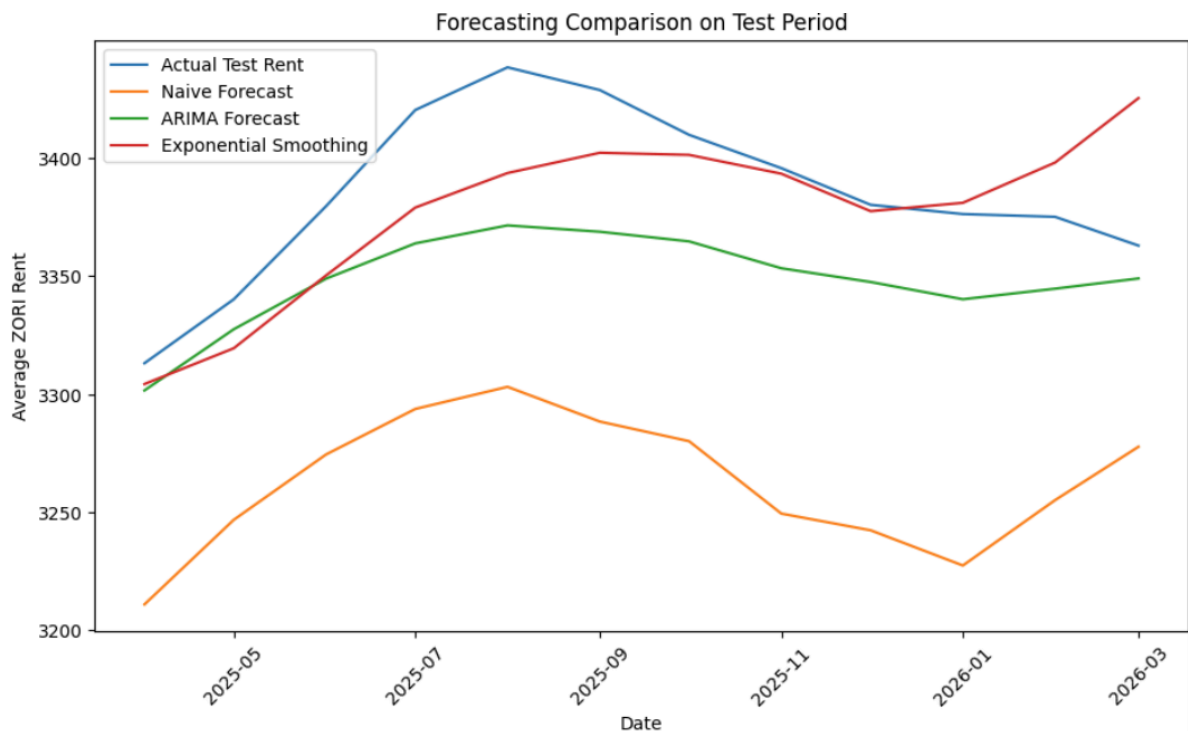
8. Smoothing

Exponential Smoothing performs best because NYC average rent changes smoothly over time, and this method gives more weight to recent observations. Since recent rent levels are highly informative, exponential smoothing **adapts better to the test period**.

Exponential Smoothing performs best, ARIMA performs second best, and the Naive Forecast is the weakest baseline.

	Model	MAE	RMSE
0	Naive Forecast	122.615109	124.307401
1	ARIMA	36.591826	40.664713
2	Exponential Smoothing	22.944937	29.334595

In the test-period comparison plot, **Exponential Smoothing** follows the actual rent line **most closely**, especially around late 2025 and early 2026, while Naive Forecast stays too low and **ARIMA is moderately accurate** but smoother.



VII. Conclusion

- Main finding:

The **EDA** shows that NYC rent has **strong location differences** and clear **time-series patterns**. Consistent with this, **historical rent values** are the strongest predictors of future NYC rent. Because rent changes gradually over time, current and lagged rent variables are highly useful for prediction.

- EDA finding:

The EDA shows clear **spatial and temporal patterns** in NYC rent. **Manhattan** and **LIC** have much higher rent levels, while Queens, the Bronx, and Staten Island are more concentrated in lower rent ranges. Rent growth also shows a sharp COVID-era disruption followed by a strong rebound after 2021.

- Best regression model:

Linear Regression performs best, with an MAE of about \$34 and an R^2 of about 0.998. This suggests that next-month rent can be predicted very accurately using recent rent trends.

- Classification result:

Logistic Regression performs better than KNN Classification for predicting whether a ZIP-code-month will be high-rent, with an accuracy of about 96.8%.

- Forecasting result:

Exponential Smoothing performs best among the time-series models, outperforming both ARIMA and the naive baseline. This shows that recent rent observations are especially important for short-term forecasting.

VIII. Limitations and Future Work

One major limitation of this project comes from the ZORI data itself. Although ZORI is useful for studying general rent trends, it is still an aggregated ZIP-code-level rent index rather than individual apartment listing data. Therefore, our model cannot capture apartment-level features such as unit size, floor level, amenities, building type, or exact address. These factors can create large rent differences even within the same ZIP code. This is especially important for study-away students, because they often prefer newer or higher-end apartments with better amenities and transportation access, whose actual rents may be much higher than the ZORI value predicted by our model.

Another limitation is that our model mainly tests whether historical rent trends can predict future rent, but it does not fully explain why rent changes. External factors such as subway access, neighborhood income level, new housing supply, and broker fees are not directly included. Also, the forecasting extension estimates NYC average future rent rather than the exact rent of a specific apartment, building, or ZIP code. For future work, we could combine ZORI with detailed rental listing data and neighborhood-level variables, and build ZIP-code-specific forecasts to make the model more useful for renters interested in particular neighborhoods.

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